

HashCode

Data Structures
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```
public int hashCode() {  
    // calculate some int  
}
```

HashCode Contract:

- ① HashCode should recalculate the same result on the same Object during a single execution of your program
- ★ ② If two Objects are .equals → true, then they should have the same HashCode
if equals methods are based on X, Y, Z fields then use the same fields in the hashCode method
- ③ Two objects with the same hashCode result they do not have to be equal

```
Ex: public class CellPhone{
```

```
    private String carrier;  
    private String phoneNumber;  
    private double versionNumber;
```

```
    ...  
    public String toString() {  
        return ...
```

```
    }
```

```
    public boolean equals(CellPhone other) {  
        return this.phoneNumber.equals(other.phoneNumber);
```

```
    }
```

```
    public int hashCode() {
```

```
        // must use phone # here
```

```
    }
```